

Undoped Ruthenium Oxide as a Stable Catalyst for the Acidic Oxygen Evolution Reaction

Corresponding Author: Professor Zongping Shao

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors investigated the catalytic performance of two commercial RuO₂ catalysts in a real PEMWE cell. While it is indeed surprising that pure RuO₂ can remain stable at a large current density ($>1 \text{ A cm}^{-2}$) for hundreds of hours, the primary contribution of this work lies in the proposed degradation mechanism. The authors suggest that proton intercalation plays a key role in the poor stability of Ru oxides during the OER. However, despite attempts to demonstrate this process, the evidence presented is unclear and lacks solidity. Therefore, I cannot recommend the acceptance of this article. My specific concerns are outlined below:

1. In line 187, the authors state that "... the coordination of water in h-RuOx brings about a lower Ru valence state and more oxygen vacancies." However, there is no direct evidence provided for the coordination of water in h-RuOx. Since this conclusion is central to the proposed mechanism, detailed information about the coordination of water should be included.
2. The authors claim that h-RuOx has a higher specific capacitance. However, Figure 4a shows that the specific capacitance of h-RuOx is lower than that of anh-RuOx.
3. The authors used the CV curve in the non-Faradaic region to confirm proton intercalation. However, the specific surface area or electrochemical surface area differs between h-RuOx and anh-RuOx. This difference could also account for the observed trend in the CV profile.
4. I am curious about the generation of O₂ during the thermal treatment of h-RuO₂. The release of O₂ is generally accompanied by the release of H₂O. What is the reaction equation? Is h-RuO₂ or anh-RuO₂ reduced to metallic Ru after thermal treatment in Ar?
5. In line 305, the authors state that the results "...reveal a thorough distribution of coordinated water throughout the bulk phase of h-RuOx." However, the TOF-SIMS results cannot exclude the possibility that only OH is distributed throughout the bulk phase while the coordinated water is present only on the surface.
6. The stability of commercial RuO₂ is surprising. To enable other research groups to replicate this result, detailed information regarding the assembly of the PEMWE setup and the MEA electrode should be provided.
7. Even for heat-treated hydrous RuO₂, the stability at high current density is excellent. These results are somewhat in contrast to previous reports, which showed that commercial RuO₂ was not stable even at low current density. To demonstrate the reproducibility of these findings, error bars should be added to the stability measurements, using at least three independent samples.

Reviewer #2

(Remarks to the Author)

Jiayi Tang et al. have reported on the observed variations in the stability of Ru oxides during OER, offering an in-depth analysis of mechanisms beyond lattice oxygen participation and Ru site over-oxidation, both of which have been recognized to contribute to Ru oxide anode degradation, but remains controversial. This work provides new understandings of the controversial results for the stability of Ru oxides in acidic OER, as reported in the literature. The findings have the potential to bring new insights into Ru oxide catalyst and catalyst layer design, extending the focus beyond doping to catalyst surface modification. I would recommend this work for publication once the following comments have been addressed.

1. All the abbreviations need and only need to be defined at their first appearance in the text.
2. In Figure 1, the authors present the stability tests of Ru oxides and Ir oxide under both PEMWE and RDE conditions.

Could the presence of sulfuric acid impact the stability of RuO₂?

3. The stability of RuO₂ anodes after thermal treatments was studied, but a comparison of polarization curves was not provided. Could thermal treatment affect the catalyst's activity?

4. In Figure 3a, Ru K-edge XANES spectra, an enlarged comparison of the photo energy at the normalized absorption of 0.6 also needs to be provided.

5. It is important to clarify the role and amount of the additional styrene-butadiene rubber (SBR) binder introduced to the catalyst layer.

6. The authors concluded that, in addition to metal cation doping, surface modification of RuO₂ could be an effective approach to achieving long-term stability of RuO₂ anodes for acidic OER in PEMWEs. Can the stability of RuO₂ be guaranteed by surface modification alone?

Reviewer #3

(Remarks to the Author)

The authors reported the use of ruthenium oxide for acidic OER in real PEMWE conditions. While Ru oxide doping has garnered significant research interest due to its potential to enhance stability, the underlying mechanisms remain insufficiently understood. This work offers a unique and novel perspective by demonstrating that undoped Ru oxide can achieve considerable stability under PEMWE operation, even at high current densities. The manuscript is well-prepared, with a thorough analysis of the proposed mechanism. Further clarification of the results discussion is recommended before considering the manuscript for publication.

1. Ru concentration was examined in the anolyte after filtration. The authors should give the calculated total concentration of Ru if the anode catalyst were to fully dissolve.

2. Please define the precious metal loadings to one decimal place when compare the electrolysis performance in PEMWE

3. Can the authors explain why anh-RuO₂ after thermal treatment conducted at 300 °C was chosen for stability evaluation? Would increasing the thermal treatment temperature further improve the stability of anh-RuO₂?

4. The Ru valence state and lattice structure remain stable after the thermal treatments conducted for the anh-RuO₂. The reasons behind this phenomenon need to be clarified.

5. Why was the stability of anh-RuO₂-300 tested for only 130 hours? Is it still stable after this period?

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

1. The evidence for proton intercalation remains unclear. The authors attribute differences in specific capacitance to this intercalation—what is the rationale for this? Additionally, the use of mass-specific capacitance for comparison is problematic. Intrinsic activity should be assessed based on ECSA, which indicates the exposure of active sites. Mass-specific capacitance assumes all sites are available for both catalysts, which is not suitable for mechanism analysis.

binding force between the membrane and catalyst. Is hot-pressing employed in this work? If so, it would be beneficial to include detailed information.

Reviewer #2

(Remarks to the Author)

I have carefully read the authors' response letter as well as the revised manuscript and Supplementary Information. I found that the authors have well-answered all my previous comments/suggestions and revised the paper accordingly. In addition, I have read the comments and response for two other reviewers, and I believe the authors have well addressed these comments as well. I thus recommend acceptance of this work without further revisions.

Reviewer #3

(Remarks to the Author)

The raised issues have been well solved by a series of additional experiments and explanation. The quality of this work has been significantly improved. Thus, it can be accepted without further modification.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I am now satisfied with the revisions authors made

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Point-to-Point Responses to Reviewers' Comments and Suggestions

Reviewer #1 (Remarks to the Author):

The authors investigated the catalytic performance of two commercial RuO₂ catalysts in a real PEMWE cell. While it is indeed surprising that pure RuO₂ can remain stable at a large current density ($>1 \text{ A cm}^{-2}$) for hundreds of hours, the primary contribution of this work lies in the proposed degradation mechanism. The authors suggest that proton intercalation plays a key role in the poor stability of Ru oxides during the OER. However, despite attempts to demonstrate this process, the evidence presented is unclear and lacks solidity. Therefore, I cannot recommend the acceptance of this article. My specific concerns are outlined below:

Response: We sincerely thank the reviewer for your thorough assessment of our manuscript and for highlighting the concerns regarding the clarity and strength of the evidence presented.

stability achieved by pure RuO₂ and offer deeper insights into the potential underlying mechanisms. We hope these revisions address the reviewer's concerns and clarify the robustness of our findings. We are grateful for the opportunity to submit a revised version and would greatly appreciate the reviewer's reconsideration of our manuscript for publication in Nature Communications. We strongly believe that this work may offer valuable insights into the future design of Ru oxide-based catalysts for OER, with simplified procedures and applicability in real PEMWE conditions. We are also willing to address any further comments from the reviewer on this manuscript.

1. In line 187, the authors state that "... the coordination of water in h-RuO_x brings about a lower Ru valence state and more oxygen vacancies." However, there is no direct evidence provided for the coordination of water in h-RuO_x. Since this conclusion is central to the proposed mechanism, detailed information about the coordination of water should be included.

Response: Thank you very much to the reviewer for your valuable comments. We carefully re-examined the original text in line 187 and have made revisions accordingly. Initially, we made an assumption that proton participation plays a crucial role in causing the instability of Ru oxide in OER. This assumption was then primarily supported by the experimental data and characterization results mainly presented in Figures 4 and 5. Thus, it could be premature to conclude the role of water coordination in advance. **In response, we have revised the text to emphasize only the lower Ru valence state**, which is clearly evidenced by the Ru K-edge XANES (Figure 2d) spectra, and O K-edge XAS (Figure 2f).

2. The authors claim that h-RuO_x has a higher specific capacitance. However, Figure 4a shows that the specific capacitance of h-RuO_x is lower than that of anh-RuO_x.

Response: We appreciate the reviewer's careful attention to detail in reviewing the plot. We sincerely apologize for the labelling error in Figure 4a, where we mistakenly reversed h-RuO_x and anh-RuO₂. **The figure has now been corrected, and the calculated capacitance values are provided in the supplementary figure S12.** As shown below, the specific capacitance of the h-RuO_x is significantly higher than that of the anh-RuO₂.

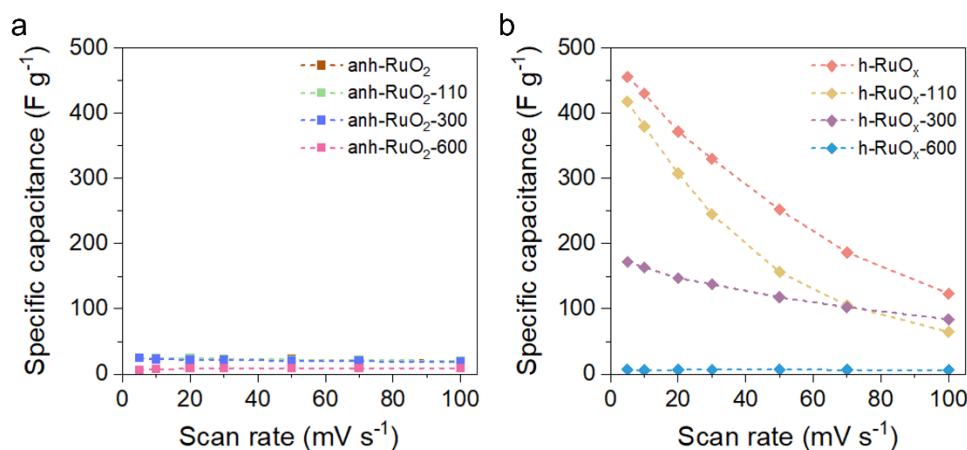


Figure S12. Specific capacitance as a function of scan rates of **a**, the anh-RuO₂ and **b**, the h-RuO_x before and after the thermal treatments.

3. The authors used the CV curve in the non-Faradaic region to confirm proton intercalation. However, the specific surface area or electrochemical surface area differs between h-RuO_x and anh-RuO_x. This difference could also account for the observed trend in the CV profile.

Response: Thank you very much to the reviewer for raising this question. We have carefully considered it. We agree with the reviewer that the specific surface area or electrochemical surface area can differ between h-RuO_x and anh-RuO₂. **Therefore, we have used mass specific capacitance in our calculations.** Since this work focuses on catalyst performance under practical PEMWE conditions, where the mass loading of the catalyst is typically controlled during fabrication and testing, comparing mass-specific capacitance is more appropriate than area-specific capacitance. Additionally, the CV curves in the non-Faradaic region, used to calculate the capacitance, were collected using a two-electrode system, which closely mimics actual PEMWE operating conditions. With these considerations, the differences in surface area would not affect the validity of the comparison results.

4. I am curious about the generation of O₂ during the thermal treatment of h-RuO₂. The release of O₂ is generally accompanied by the release of H₂O. What is the reaction equation? Is h-RuO₂ or anh-RuO₂ reduced to metallic Ru after thermal treatment in Ar?

Response: We sincerely appreciate the reviewer's insightful question. **We have addressed this point in the revised manuscript.** In response to the comment that the release of O₂ during the temperature-programmed thermal treatment process is generally accompanied by the release of H₂O, we would like to clarify the following: for h-RuO_x, the Ru centres are largely bound by OH instead of lattice O, during the thermal treatment, the reaction $2\text{OH} \rightarrow \text{H}_2\text{O} + \frac{1}{2} \text{O}_2$ takes place, which explains why both O₂ and H₂O signals are detected simultaneously by the mass spectroscopy (MS) during temperature programming.

The h-RuO_x after treatment in an Ar atmosphere tends to expose more unsaturated Ru sites due to the removal of OH. These sites may then oxidize when exposed to small amounts of O₂ at a certain temperature, leading to an increase in the Ru valence state. However, in the case of anh-RuO₂ treated in Ar atmosphere, no O₂ or H₂O release was observed during the temperature-programmed thermal treatment process. This suggests that the lattice O bound to Ru remains stable and is not reduced to the metallic state, even under the reducing conditions of the Ar atmosphere.

5. In line 305, the authors state that the results "...reveal a thorough distribution of coordinated water throughout the bulk phase of h-RuO_x." However, the TOF-SIMS results cannot exclude the possibility that only OH is distributed throughout the bulk phase while the coordinated water is present only on the surface.

Response: We sincerely thank the reviewer for the thoughtful question. We have addressed this point in the revised manuscript. The primary reason for utilizing TOF-SIMS as a characterization is its ability to provide qualitative depth profiling for diffusion studies of positive and negative ions, even for light ions. However, we recognize that TOF-SIMS is not a quantitative technique, thus we understand the reviewer's concern regarding the clarity of our definitions of the surface and bulk phases. From the X-Z depth profiling of TOF-SIMS, we observed a significant decrease in the OH⁻ distribution on the anh-RuO₂ particles as the bombardment depth increased, whereas the detection signal for OH⁻ on h-RuO_x particles remained consistently strong. The difference between the two samples is further distinguished by the intensity of OH⁻ and O²⁻ signals detected in the MS spectrum (Figure 4d).

To strengthen the conclusions, we have also combined XRD and synchrotron radiation characterizations (Figure 2) to analyze the Ru valence states of surface and subsurface of the two catalysts, demonstrating the surface coordination difference of Ru. Notably, the conclusion of no coordinated OH⁻ is presented in the bulk phase of anh-RuO₂ can be further supported by the programmed temperature desorption with MS signal detection (Figure 4b,c). Based on these combined characterizations, we can thus clarify the difference between the two Ru oxides and have made adjustments in the revised manuscript accordingly.

6. The stability of commercial RuO₂ is surprising. To enable other research groups to replicate this result, detailed information regarding the assembly of the PEMWE setup and the MEA electrode should be provided.

Response: We sincerely thank the reviewer for the valuable comment on the reproducibility of the results. In response to the concern raised, we **have provided a detailed description of the MEA preparation process and the PEMWE setup in the revised experimental section to facilitate replication by other research groups.** It is also important to note that, unlike the typical three-electrode system requirements for catalyst activity testing, we have spent the past few years extensively investigating the formation of catalyst layers, and the factors that contribute to the exceptional performance of PEMWE (Small Sci. **2023**, 3, 2300055; Energy Environ. Sci. **2024**, 10.1039/D4EE02722A). These findings give us confidence that the electronic conductivity of the catalyst layer, along with the electrolyzer's ohmic resistance, are key determinants of the overall efficiency and reproducibility of the excellent performance observed in our system. The specific details of the MEA fabrication and the testing procedures are now clearly outlined in the revised manuscript.

Preparation of MEA: "...The anode catalyst layers were formed onto the NR212 membrane first by spraying coating with the precious metal loading measured to be 1.1 mg cm^{-2} . First, the anode oxide catalyst was mixed with D521 ionomer dispersion at a weight ratio 10 wt.% for preparing the catalyst ink, and isopropyl alcohol/water solution (2:1 v/v) was used as the solvent for catalyst dispersion. At specific circumstance, the SBR was added as additional binder at an oxide/SBR ratio of 10 wt.%. The catalyst ink was treated under ultrasonic water bath for 1 h with the temperature controlled at 5°C during this process. The spraying process was conducted using a manual spray gun (HD131, 0.2-micron diameter), with the hot plate temperature maintained at 98°C throughout the procedure to ensure uniform deposition. The formation of the cathode catalyst layers based on Pt/C catalyst (HiSPEC 8000, nominally 50 wt.%, Alfa Aesar) on the other side of the membrane followed the similar procedures. An ionomer to carbon (I/C) ratio of 60 wt.% was applied in the preparation of cathode catalyst ink. The catalyst-coated membrane (CCM) with a demarcated area of $2 \times 2 \text{ cm}^2$ covered by uniform catalyst layers was dried in an oven heated up to 80°C for 1 h before making it ready to use..."

Fabrication of PEMWE: "...The fabrication of the PEMWE electrolyzer was carried out as follows: the as-prepared CCM was coupled with Toray 090 carbon paper (5 wt.% wet-proof, Fuel Cell Store) as the cathode gas diffusion layer, and platinized titanium fiber (0.27 mm, Fuel Cell Store) as the anode, with both components cut to $2 \times 2 \text{ cm}^2$ to form the membrane electrode assembly. PTFE gaskets, 0.3 mm thick, were used for sealing. The single-cell electrolyzer was assembled by applying 2.5 Nm of torque to fasten the bolts..."

PEMWE Tests: "...deionized water heated to 80°C was supplied to the anode side at a flow rate of 80 mL min^{-1} , with both the electrolyzer and circulating water temperatures stabilized at 80°C . The polarization curves were obtained on a Keysight DC power supply (30V, 20A) by incrementally stepping the voltage from 1.2 V to 1.8 V at intervals of 10 mV. The corresponding currents were recorded after 10 s of equilibrium at each voltage point..."

7. Even for heat-treated hydrous RuO_2 , the stability at high current density is excellent. These results are somewhat in contrast to previous reports, which showed that commercial RuO_2 was not stable even at low current density. To demonstrate the reproducibility of these findings, error bars should be added to the stability measurements, using at least three independent samples.

Response: Thank you very much for the reviewer's insightful comment. We also noted that previous reports have shown instability of commercial RuO_2 at even low current densities. We intended to investigate this phenomenon, however, during which, we occasionally find the huge

difference in stability between the hydrous RuO_2 and its heat-treated forms. Thus, we thought about demonstrating this unique finding in this work. The reviewer's suggestion is extremely helpful in further demonstrating the reproducibility of our work. **In response, we repeated the stability tests three times using h-RuO_x-300 (prepared by treating the hydrous RuO_2 at 300 °C in air for 2 h) as the anode.** The results of these reproducible experiments, which are included as supplementary materials, confirm the stability improvements we aim to highlight in this work. In particular, this does not conflict with previously reported works, where doped RuO_2 catalysts exhibited stability after synthesis under calcination conditions at or above 300 °C (Table S2). In addition to the role of doping, our findings suggest that heat treatment alone, by removing bound water and reducing the unsaturated coordination of Ru, plays a critical role in enhancing the stability.

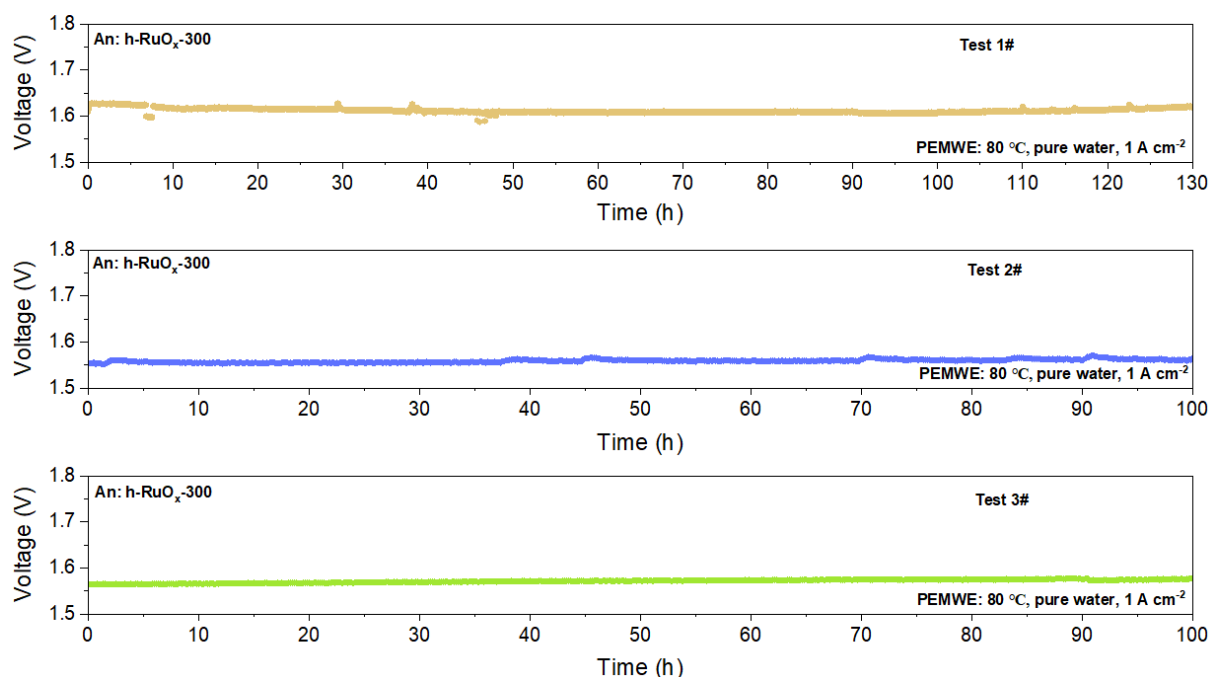


Figure S19. Repeated PEMWE stability testing with the h-RuO_x-300 anode for operating at 1 A cm⁻².

Reviewer #2 (Remarks to the Author):

Jiayi Tang et al. have reported on the observed variations in the stability of Ru oxides during OER, offering an in-depth analysis of mechanisms beyond lattice oxygen participation and Ru site over-oxidation, both of which have been recognized to contribute to Ru oxide anode degradation, but remains controversial. This work provides new understandings of the controversial results for the stability of Ru oxides in acidic OER, as reported in the literature. The findings have the potential to bring new insights into Ru oxide catalyst and catalyst layer design, extending the focus beyond doping to catalyst surface modification. I would recommend this work for publication once the following comments have been addressed.

Response: We appreciate the reviewer for recommending our manuscript! We have carefully revised the manuscript by addressing all the following comments. Point-by-point responses are listed below and the revised context in the manuscript is explained in detail as follows.

1. All the abbreviations need and only need to be defined at their first appearance in the text.

Response: Thank you for pointing this out. We have thoroughly reviewed the entire manuscript and carefully double-checked all uses of abbreviations. In particular, we have removed any duplicate occurrences of abbreviations in the figure captions to improve readability.

2. In Figure 1, the authors present the stability tests of Ru oxides and Ir oxide under both PEMWE and RDE conditions. Could the presence of sulfuric acid impact the stability of RuO₂?

Response: Thank you very much for the reviewer's question. We have added further explanation to address this point. In the PEMWE and RDE tests, acidic conditions for catalytic reactions are provided by a solid acid (ionomer) and liquid acid (0.5 M H₂SO₄), respectively. We conducted stability tests under PEMWE conditions using both pure water and a 0.5 M H₂SO₄ circulating electrolyte, observing no significant differences in stability—anh-RuO₂ still demonstrated excellent stability. It is also noteworthy that when ionomers are fully and uniformly distributed in the PEMWE electrodes, the necessary proton conductivity is adequately provided. Therefore, additional 0.5 M H₂SO₄ does not enhance any activity as well, as the reaction interface in electrode is formed by the contact between the catalyst and the ionomer. The use of either 0.5 M H₂SO₄ or pure water as a feedstock does not directly impact this interface.

3. The stability of RuO₂ anodes after thermal treatments was studied, but a comparison of polarization curves was not provided. Could thermal treatment affect the catalyst's activity?

Response: We thank the reviewer for raising this valuable question out. Accordingly, we have supplemented additional data regarding the PEMWE performance with the h-RuO_x anodes after thermal treatments at 300 °C and 600 °C in the revised manuscript. We also provided a comparison with both the anh-RuO₂ and untreated h-RuO_x. The results indicate that the h-RuO_x-300, after treatment at 300 °C, still exhibit higher activity than anh-RuO₂, and perform even better than the untreated h-RuO_x. This is likely due to the retention of some oxygen vacancies after the removal of binding –OH. Following heat treatment at 600 °C, the performance with the h-RuO_x-600 decreases much, which could be attributed to the growth of catalyst particles and the improved crystallinity with a reduction of oxygen vacancies. This balance between activity and stability also supports the proposed stability mechanism effectively.

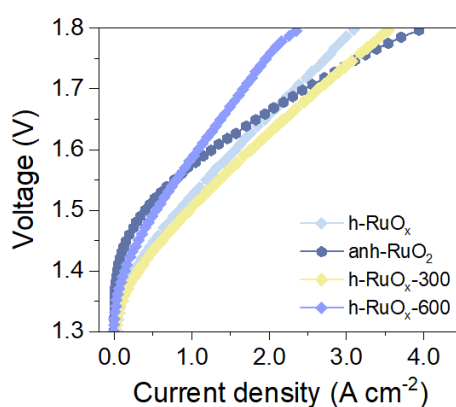


Figure S18. The PEMWE performance curves of the h-RuO_x-300 and h-RuO_x-600 catalysts after the thermal treatments conducted under 300 °C and 600 °C, respectively, compared with the h-RuO_x anode and anh-RuO₂ anode.

4. In Figure 3a, Ru K-edge XANES spectra, an enlarged comparison of the photo energy at the normalized absorption of 0.6 also needs to be provided.

Response: We sincerely thank the reviewer for this comment. We have revised **Figure 3a** as recommended, and we have included an enlarged comparison of the photon energy at a normalized absorption of 0.6. As shown in Figure 3a and 3b, the Ru K-edge XANES and EXAFS spectra confirmed that both the Ru valence state and local structure remain stable in the anh-RuO₂ anode after stability tests.

5. It is important to clarify the role and amount of the additional styrene-butadiene rubber (SBR) binder introduced to the catalyst layer.

Response: We appreciate the reviewer for the valuable suggestion to help with the reproducibility of the results in this work. In response to the concern raised, we have provided a detailed description of the catalyst layer formation process and the PEMWE setup **in the revised experimental section**: “....At specific circumstance, the styrene-butadiene rubber (SBR) was added as additional binder at an oxide/SBR ratio of 10 wt.%....”

6. The authors concluded that, in addition to metal cation doping, surface modification of RuO₂ could be an effective approach to achieving long-term stability of RuO₂ anodes for acidic OER in PEMWEs. Can the stability of RuO₂ be guaranteed by surface modification alone?

Response: We sincerely thank the reviewer for this thought-provoking question, which has inspired new directions for our future work. Building on the insights gained from this study, we believe that in developing next-generation Ru oxide catalysts to replace Ir in PEMWEs, stability considerations should address two key aspects: **the stability of the catalytic sites and the stability of the reaction interface**. Since the electrode reaction takes place on the catalyst surface, surface modification emerges as a critical strategy in tuning both the aspects. However, an effective surface modification strategy needs to both ensure stable Ru electronic structure, also eliminate the reaction interface changes during the OER. Both are indispensable for optimal stability. From an electronic structure perspective, we believe that doping or adjusting surface coordination can still enhance surface stability as highlighted in many works today. Additionally, to ensure interface stability, introducing binders that help prevent catalyst detachment may be an effective approach.

Reviewer #3 (Remarks to the Author):

The authors reported the use of ruthenium oxide for acidic OER in real PEMWE conditions. While Ru oxide doping has garnered significant research interest due to its potential to enhance stability, the underlying mechanisms remain insufficiently understood. This work offers a unique and novel perspective by demonstrating that undoped Ru oxide can achieve considerable stability under PEMWE operation, even at high current densities. The manuscript is well-prepared, with a thorough analysis of the proposed mechanism. Further clarification of the results discussion is recommended before considering the manuscript for publication.

Response: We are much appreciated for the reviewer's comments and valuable suggestions to making a major revision of our manuscript. The explanations and discussions as required have been supplemented to the revised version of manuscript and point-by-point responses are listed as follows.

1. Ru concentration was examined in the anolyte after filtration. The authors should give the calculated total concentration of Ru if the anode catalyst were to fully dissolve.

Response: We are much appreciated for the reviewer's thoughtful comments. Considering that the loading of precious metal Ru in the anode catalyst layer is measured as 1.1 mg cm⁻² across an effective electrode area of 4 cm², and the water volume in the anode circulating container is consistently maintained at 300 mL through an automatic water replenishment device to ensure stable operation. If the Ru were to completely dissolve, the resulting concentration in the solution would be ~ 13.3 ppm. However, our ICP measurements consistently showed that the Ru content in the solution is actually at the ppb level. This suggests that the loss of Ru due to

dissolution is extremely minimal and nearly undetectable. This finding reinforces the conclusion that metal dissolution is not the major cause for the performance degradation.

2. Please define the precious metal loadings to one decimal place when compare the electrolysis performance in PEMWE

Response: Thanks for the reviewer's suggestion. Accordingly, we have made additional edits to the revised manuscript. We employed a weighing method to assess the oxide catalyst-coated membrane prepared by spraying coating for 3 times, and the resulted precious metal loadings are measured to be 1.04, 1.12, and 1.10 mg cm⁻², respectively, which confirms that the precious metal loading should be reasonably controlled at approximately 1.1 mg cm⁻² with one decimal place. This consistency in loading is crucial for ensuring the performance repeatability of the catalyst in our experiments.

3. Can the authors explain why anh-RuO₂ after thermal treatment conducted at 300 °C was chosen for stability evaluation? Would increasing the thermal treatment temperature further improve the stability of anh-RuO₂?

Response: We sincerely appreciate the reviewer's insightful comment. We selected a heat treatment temperature of 300 °C to investigate the stability improvements, primarily because the synthesis of doped Ru oxide catalysts reported in the literature is frequently conducted within the range of 300 to 350 °C (as shown in Table S2). We believe that this temperature represents a suitable condition for studying the stability of undoped Ru oxide under PEMWE operating conditions, thereby providing valuable insights into the stability mechanisms we propose.

Regarding the reviewer's inquiry about the potential effects of increasing the thermal treatment temperature on stability, we think that higher temperatures can enhance particle crystallinity and reduce unsaturated coordination within the bulk phase. These changes may stabilize lattice oxygen, thereby improving stability. However, we also recognize that elevating the temperature can lead to particle growth, which might reduce the active surface area of the catalyst and subsequently diminish its overall activity. This relationship is illustrated by the polarization curves of PEMWE with Ru oxide anodes obtained at various heat treatment temperatures, as depicted in the supplemented Figure S18.

Taking these factors into consideration, we used 300 °C as an appropriate treatment temperature for evaluating the stability improvements.

4. The Ru valence state and lattice structure remain stable after the thermal treatments conducted for the anh-RuO₂. The reasons behind this phenomenon need to be clarified.

Response: Many thanks for the valuable suggestion. We have therefore supplemented the explanation to this observed phenomenon in the revised manuscript. "...The Ru valence state and lattice structure remain extremely stable after the same thermal treatments conducted for the anh-RuO₂ (Fig. S13 and S14), indicating that there is no chemically bonded -OH presented

with the anh-RuO₂ leaving a CUS-free bulk phase. The results are also in good consistency of the TPD profiles detected from the MS.”

5. Why was the stability of anh-RuO₂-300 tested for only 130 hours? Is it still stable after this period?

Response: Thank you very much for raising this question. To make it clear, given that untreated h-RuO_x failed within just 1 min, thus we have conducted the stability verification of h-RuO_x-300 for 130 and more hours, we see this stability improvement—over 6000 times—is significant to demonstrate the proposed mechanism. To ensure the reliability of our results, we also repeated the stability test for 3 times by using the h-RuO_x-300 anode, as illustrated in the supplemented Figure S19.

Furthermore, we anticipate that longer testing durations may witness a slow decay in stability, similar to what has been observed for anh-RuO₂. We have highlighted in the revised manuscript that this gradual decay is likely associated with reaction interface factors and proposed that it can be addressed through subsequent surface modifications.

Point-to-Point Responses to Reviewers' Comments and Suggestions

Reviewer #1 (Remarks to the Author):

The authors have partially addressed concerns, but the evidence and theoretical calculations for their proposed mechanism remain weak.

Response: We sincerely thank the reviewer for your time and efforts in helping us enhance this work. We have carefully addressed each of the concerns regarding the evidence and theoretical calculations and have added supplemental experiments to support the proposed mechanism. We are fully committed to continuing improvements and addressing any remaining concerns to ensure the quality of this work meets the reviewers' expectations.

1.The evidence for proton intercalation remains unclear. The authors attribute differences in specific capacitance to this intercalation—what is the rationale for this? Additionally, the use of mass-specific capacitance for comparison is problematic. Intrinsic activity should be assessed based on ECSA, which indicates the exposure of active sites. Mass-specific capacitance assumes all sites are available for both catalysts, which is not suitable for mechanism analysis.

Response: We sincerely thank the reviewer for your insightful concerns, which have been instrumental in elevating the quality of our work. Following the comments, we have provided additional explanations and supplementary data to support our findings.

To address the reviewer's question on the theoretical basis of using specific capacitance to characterize proton intercalation in the OER process, we found in many previous works (*Electrochimica Acta*, 49(2), pp.313-320; *Electrochemistry communications*, 8(9), pp.1539-1543; *Chemical Science*, 10(22), pp.5656-5666.), the pseudocapacitive behaviours of Ru oxides were highlighted. Specifically, this behaviour encompasses proton redox and intercalation onto the Ru oxide surface, which differs significantly between various Ru oxide materials. For instance, the two Ru oxides we selected to investigate in this work—both commercially available h-RuO_x, commonly used in supercapacitors, and anh-RuO₂, favoured as electrocatalysts, exhibit distinct differences in both activity and stability.

During the electrode reaction in PEMWE, no matter whether the OER process follows the LOM or the AEM mechanism, the –OH is actively bounded to the Ru site, followed by the proton and electron transfer. When we perform capacitance measurement under acidic conditions, proton participation is isolated by preventing OER, thereby independently reflecting the proton redox and intercalation properties on the Ru oxide. The equations are listed as follows:



In our ongoing work on Ru-based catalyst design, we continue to find this method valuable, as proton participation likely has a notable impact on electrode performance, especially in PEMWE applications.

Following the suggestion, we have supplemented the intrinsic activity comparison based on ECSA. The ECSA of the Ru oxides was estimated from the electrochemical double-layer capacitance (Cdl) measurements under RDE conditions as shown below (supplemented as **Figure S23**), and the specific capacitance (Cs) used for calculation is 0.035 mF cm⁻². The as-calculated ECSA for h-RuO_x is 25.2 m² g⁻¹, and for anh-RuO₂ is 11.7 m² g⁻¹. The intrinsic activity is then compared in supplementary **Figure S24**.

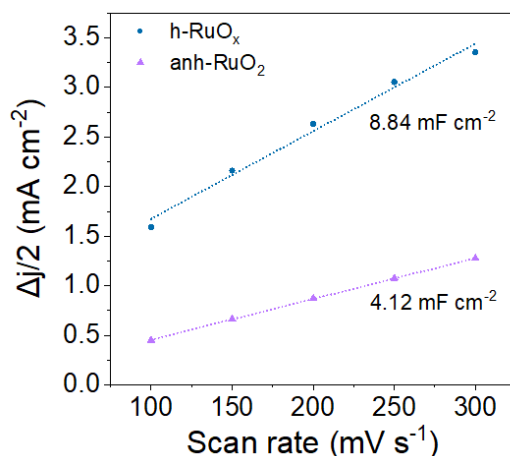


Figure S23. The electrochemical double-layer capacitance obtained from the liner fitting of $\Delta j/2$ against scan rates.

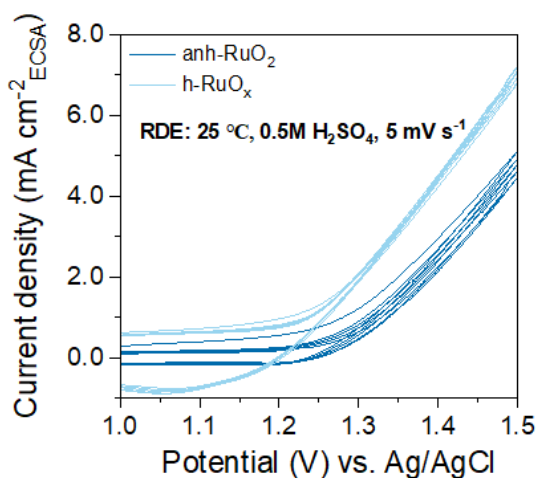
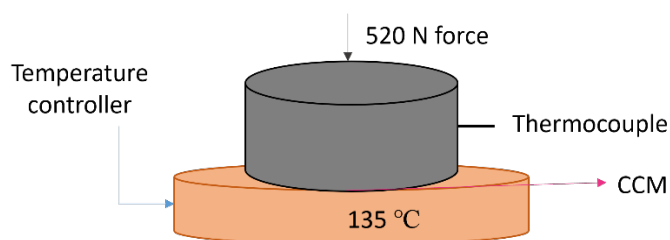


Figure S24. A comparison of the cyclic voltammetry curves from the anh-RuO₂, and h-RuO_x with the current density normalized to ECSA.

Following the reviewer's suggestion, we have also included a comparison of area-specific capacitance calculated based on ECSA, and the corresponding changes have been made to **Figure 4a** and the description in the revised manuscript.

3. The hot-pressing process is widely recognized as an effective method to uniformly redistribute ionomers and enhance the binding force between the membrane and catalyst. Is hot-pressing employed in this work? If so, it would be beneficial to include detailed information.

Response: We sincerely thank the reviewer for raising this critical point regarding the role of hot pressing in redistributing ionomer and enhancing membrane-catalyst binding. We deeply value the reviewer's expertise in this area and strongly agree with the perspective, as we have also focused considerable efforts on optimizing our PEMWE assembly conditions. Although our lab setup currently lacks a dedicated hot-pressing machine to precisely control this process, we have developed an alternative setup, as shown below, to closely approximate the required hot pressing conditions with the available equipment.



In our hot-pressing setup for the dried CCM, a flat-surfaced iron block was heated on a hot plate, with a calibrated weight positioned on top to provide a stable applied force. The iron block and hot plate were heated to a consistent 135°C, and the CCM, covered with a cellulose-based paper layer, was placed in between. This setup exerted a total force of approximately 520 N on the CCM, translating to a pressure of about 42.25 kPa for 30 s—somewhat lower than the >100 kPa commonly reported in the literature but sufficient for consistent results in our PEMWE tests.

As the reviewer recommended, we have added these details to the revised manuscript. Alongside the pressing process, we attribute the strong PEMWE performance partly to the optimized ionomer content in the catalyst layer, set at 10 wt.%, which ensures both electronic and proton conductivity at the reaction interface. Additionally, the intrinsic activity of RuO₂, which surpasses that of IrO₂, likely contributes to enhanced electrolyzer performance. Together, these optimized parameters have allowed us to achieve stable PEMWE operation, even with slightly lower-than-standard hot-pressing conditions. We would be grateful for any further insights the reviewer might have and welcome the opportunity to discuss additional strategies for optimizing these conditions.